

Project Summary

Attracting and Retaining Undergraduates to Engineer the Built Environment through Instructional & Technological Innovation

The disciplines of Civil and Environmental Engineering (CEE) play a critical role in the development and maintenance of physical infrastructure and hence in any strategy for improving planetary habitability and sustainable use of resources. The breadth of topics covered by these disciplines and the need to understand their interactions in a physical, chemical, ecological, social and economic context pose challenges for the design of an undergraduate curriculum. The traditional, sequential, skill based undergraduate curriculum has successfully trained a cadre of engineers who have designed and built reliable physical infrastructure that society takes almost for granted. However, “engineer bashing” is not uncommon in many segments of society as this infrastructure ages, needs replacement, and unanticipated environmental and social problems created as a byproduct come to the fore. Re-engineering the \$20 trillion US physical infrastructure is now recognized as a major challenge. At the same time, undergraduate CEE enrollments have been declining, and retention of women and other minorities lag expectations.

A prevailing response to this situation is that an undergraduate engineering program cannot adequately provide the breadth and depth of coverage needed, particularly given the humanities, liberal arts and basic science requirements. Consequently, the MS is being proposed as the entry level professional degree and various mechanisms for combined BS-MS degrees, coursework only MS degrees and the like are being explored and implemented. While this conclusion and the resulting direction may be inevitable in the traditional education model, it is unclear that they constitute the best response for educating engineers as analysts and as “master integrators” or leaders in providing solutions for meeting the needs of managing a complex and changing system.

Interviews with students reveal: (1) a strong interest in the CEE subject matter, but a lack of knowledge at the college entrance level of the engineer’s role; (2) little exposure to engineering in the first two years; and (3) a sense of ill-preparedness for the work force on graduation that is often ascribed to instructional methods that are abstract, focused on general principles related to unit processes, with limited integrative application. The second item contributes negatively to undergraduate retention at schools where the student has a choice of major through the second year. An institutional response is to emphasize design content, and introduce capstone or synthesis classes in the curriculum. However, these measures are typically adopted only in junior/senior classes.

Through a partnership between Columbia University’s Departments of Civil Engineering & Engineering Mechanics and Earth & Environmental Engineering and the Columbia Center for New Media Teaching and Learning the development of a new instructional approach and curriculum to systematically address the issues raised above is proposed here. The new curriculum is anchored by a sequence of classes in the first three years that progressively expose the students to a variety of CEE problems of regional and national interest in a case study mode. These classes, designed to provide technological literacy to all majors, will use a “systems approach”. The proposed curriculum will be progressively specialized as one moves to the higher grades, offering a student the opportunity to explore a subsystem given an understanding of the larger context of the problem. The goal is to present a data driven, problem focused approach to learning that integrates material from concurrent science and humanities classes and emphasizes the interconnection between most classes in the curriculum. Spatial interactions between subsystems over different time frames will be made explicit and used to motivate the application of empirical as well as theoretical approaches to design, analysis and management of the pieces and the whole.

Information and computer technology will be used extensively to develop a virtual reality platform that will be used throughout the curriculum. Beyond the redeveloped curriculum, the development of this open simulation platform and working prototype scenarios for classroom introduction represents the major deliverable of this proposal. Spatially explicit simulators that bring together a variety of interacting infrastructure and environmental components (as in the game *SimCity*) will be used as a vehicle to introduce the broad problem context, and to bring case studies to life. Student teams will use it to explore historical data, as well as the effects of both policy and structural measures for a range of problems (e.g., natural and environmental hazards) on the long term functioning of the infrastructure, the environment and interacting social systems. In addition to exercising the system with different problems and settings, students will develop and add functional modules to the system using a high-level programming language. The inner workings of key subsystem components or processes will be accessible for stand-alone instruction. Interaction with existing numerical models, Geographical Information Systems, and statistical analysis packages will be considered as part of the design. Case studies introduced in the first year may continue to be building blocks for more detailed analyses in subsequent years.

NSF Support is sought for the development of CEE case studies and the educational simulation/gaming platform. Materials for three classes offered in the first two years will be developed using the proposed grant and matching funds. Industrial collaborators who have experience developing SIMgames have been recruited as consultants to help design and implement the open modeling platform. The materials developed under the grant will be made available to the engineering education community, and formal evaluations of changes in material, skills and knowledge imparted to and retained by students will be performed.